ENGINEERING PORTFOLIO

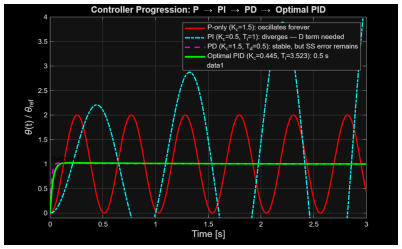
Ariadne Colina-Valeri

Hi! My name is Ariadne, and I'm a junior studying Electrical Engineering at Tufts University. I have experience in signal processing, embedded systems and machine learning, from restoring RF signals with a Wiener filter to building a real-time audio effects board and an autonomous robot. I'm looking for internships in signal processing, embedded systems and machine learning, in biotech or defense.

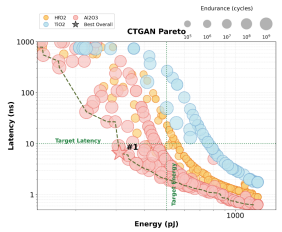


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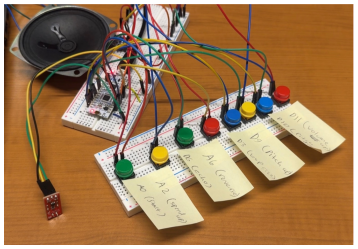
SELECTED PROJECTS



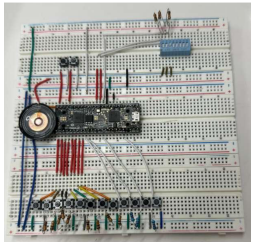
UAV Pitch Control
page 2



CTGAN for RRAM
page 3



DJ Board on an STM32
page 4



FPGA Synthesizer
page 5

PROJECT 1 | LINEAR SYSTEMS | JAN. 2026 TO MAY 2026

UAV Pitch Control and RF Signal Restoration

THE CHALLENGE

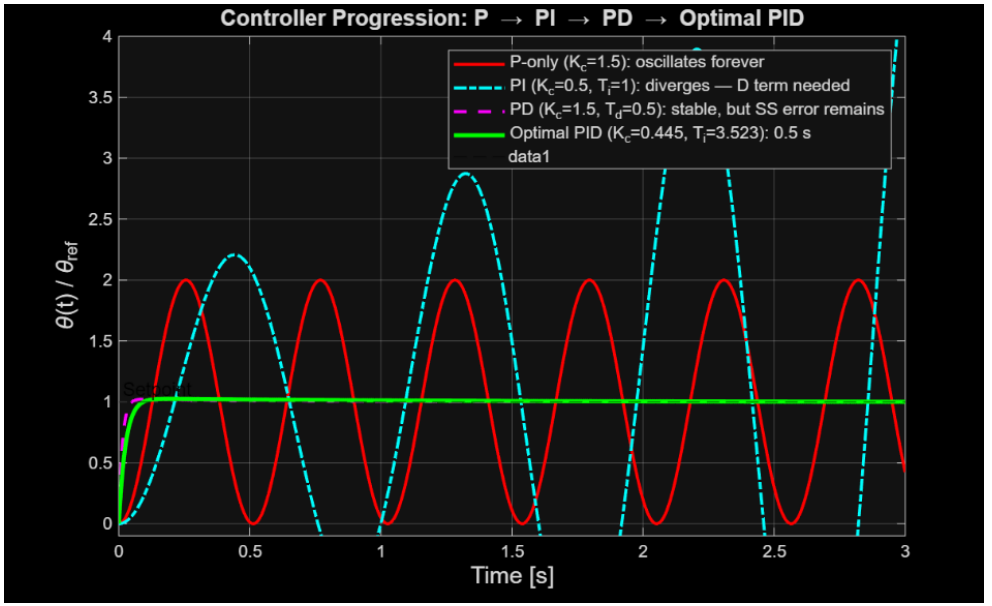
A UAV that shakes while collecting RF signals blurs them, so it has to stay level first.

WHAT I DID

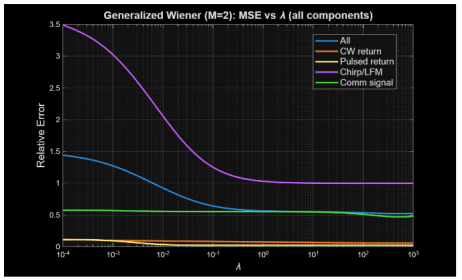
I proved with Routh-Hurwitz analysis that P and PI controllers can never stabilize it, searched 15,625 gain combinations for the best PID settings, then restored a blurred, noisy RF signal with a Wiener filter in MATLAB.

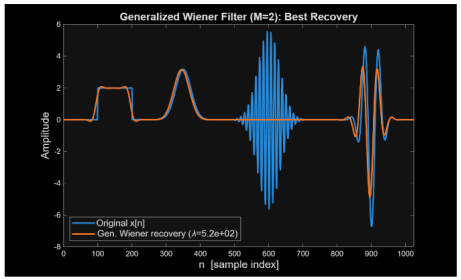
THE RESULT

A 52ms rise time and 2.6% overshoot, within real UAV specs, and 77% less error than a direct inverse.



P, PI, PD and PID controllers on the same UAV model





Wiener filter error (left) and the restored RF signal (right)

PROJECT 2 | TUFTS EMERGING CIRCUITS AND SYSTEMS LAB | SEPT. 2025 TO APRIL 2026

CTGAN Design Discovery for RRAM

THE CHALLENGE

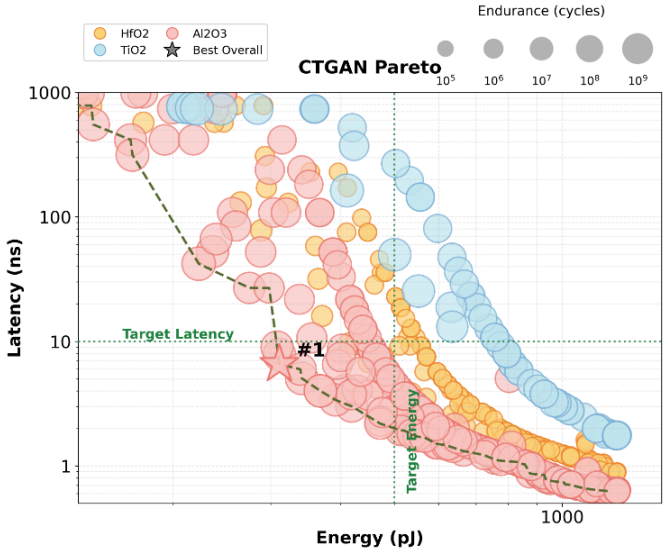
The published framework I extended used a CVAE, which averages its training data and smooths away the extreme designs where the best performance is found.

WHAT I DID

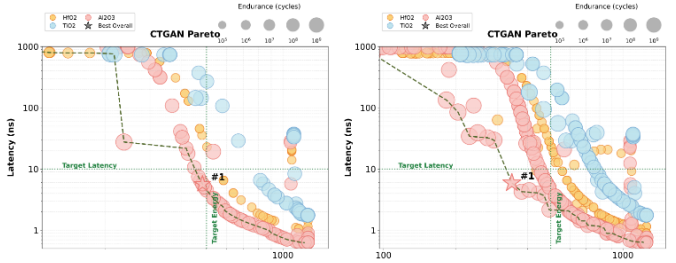
I replaced it with a CTGAN that searches those extremes more aggressively, used the framework's physics model (PINN) to reject impossible designs, and grew the training data tenfold to 5,000 samples after the model collapsed at 500.

THE RESULT

A design with 25% higher endurance and 11% lower switching energy than the published result, with slightly slower switching (6.57ns vs 4.07ns).



Final results with 5,000 training samples



The same search with 500 samples (left) and 2,500 (right)

PROJECT 3 | EMBEDDED SYSTEMS | JAN. 2026 TO MAY 2026

DJ Board on an STM32

THE CHALLENGE

Seven real-time audio effects had to fit in the STM32's 64KB of on-chip memory, and the first playback came out with a high-pitched hiss and clicks.

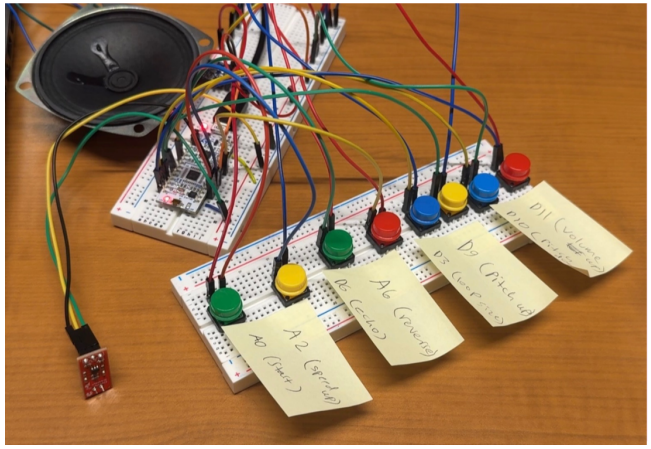
Microphone → ADC → buffer → effects → DAC → amplifier → speaker

WHAT I DID

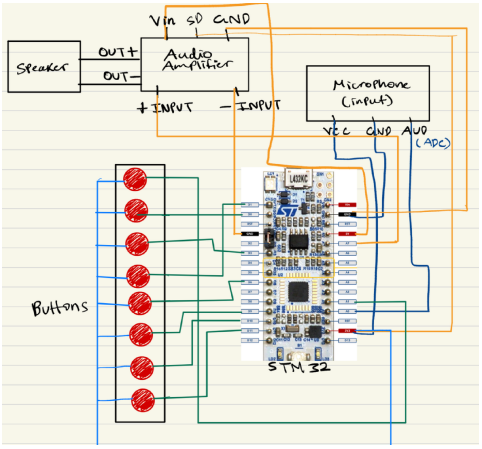
To remove the hiss, I used a 4kHz timer interrupt to record and play every sample at the same rate, and lengthened the ADC sampling time to 12.5 cycles so the microphone signal settles before each reading. The clicks came from sudden jumps in the speaker signal, so I held the output at mid-rail between clips and faded each clip in and out over 256 samples (64ms). Dropping from 8kHz to 4kHz stretched each recording from 2 to 5 seconds.

THE RESULT

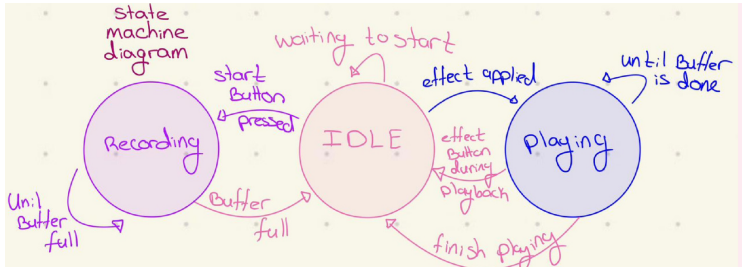
5 seconds of 12-bit audio in a 40KB buffer, all 7 effects (echo, reverse, speed, loop size, pitch up, pitch down, volume), and a playback waveform that matched the original on the oscilloscope.



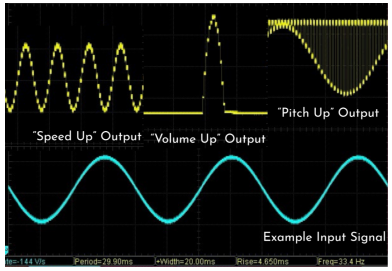
The board



Hardware: wiring diagram



Software: the state machine



Oscilloscope captures of the effects

PROJECT 4 | DIGITAL LOGIC CIRCUITS | SEPT. 2025 TO DEC. 2025

FPGA Audio Synthesizer

THE CHALLENGE

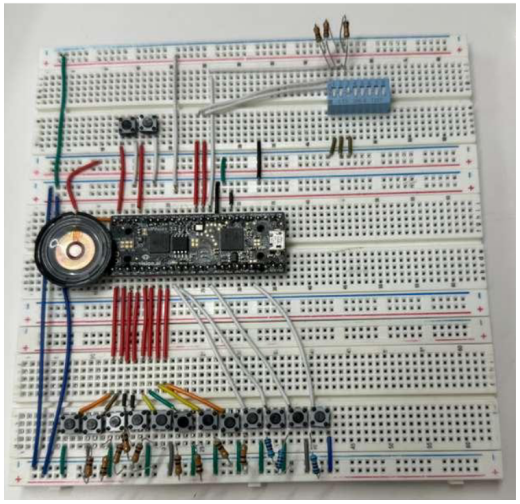
Each octave doubles a note's frequency, and my first design calculated every note by dividing on the chip, which needed over 250% of the FPGA's space.

WHAT I DID

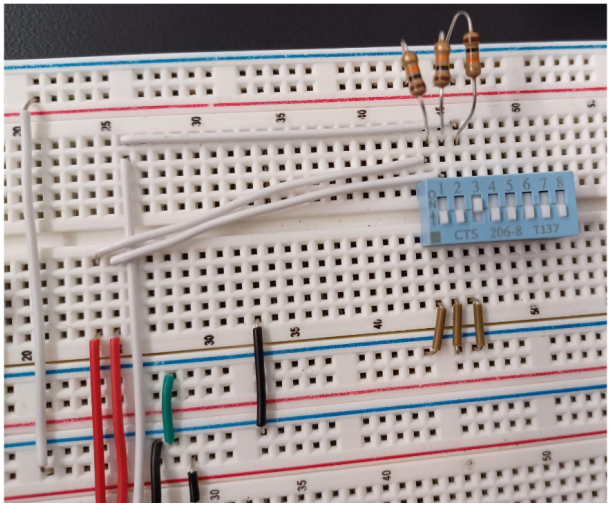
Since dividing by 2 is the same as shifting bits, I hand-calculated one octave and used bit shifts for the rest, then mixed the notes with PWM after chords distorted.

THE RESULT

A full 12-note scale across 8 octaves (16Hz to 8kHz) on a MIDI keyboard, with chords of up to 6 clear notes.



The complete board



The switch that selects the octave